

OpenCV Curriculum

Beginner → Advanced | Structured OpenCV learning roadmap

Goal: Build strong practical OpenCV skills from image fundamentals and processing through features, video, camera geometry, 3D vision, DNN inference, and performance.

LEVEL 1 — OPENCV FUNDAMENTALS

1. Introduction to OpenCV

- What is OpenCV?
- OpenCV architecture
- OpenCV modules
- Installing OpenCV
- Python + OpenCV
- cv2
- OpenCV + NumPy
- Image representation

2. Images & NumPy

- Pixels
- Image dimensions
- Width, height, channels
- Image data types
- uint8
- Image arrays
- Indexing
- Slicing
- ROI
- Copy vs view

3. Reading & Writing Images

- cv2.imread()
- cv2.imwrite()
- cv2.imshow()
- Image paths
- Image properties
- Handling invalid images

4. Display & User Interaction

- Windows
- Mouse events
- Keyboard events
- Trackbars
- Drawing callbacks

LEVEL 2 — COLOR & IMAGE MANIPULATION

5. Color Spaces

- BGR
- RGB
- Grayscale
- HSV
- LAB
- YCrCb
- Color conversion
- Color masking

6. Basic Image Operations

- Resize
- Crop
- Flip
- Rotate
- Translate
- Padding
- Image arithmetic
- Bitwise operations

7. Drawing

- Lines
- Rectangles
- Circles
- Polygons
- Text
- Bounding boxes
- Annotations

LEVEL 3 — GEOMETRIC TRANSFORMATIONS

8. Affine Transformations

- Translation
- Rotation
- Scaling
- Shearing
- Affine transformation matrix

9. Perspective Transformation

- Perspective distortion
- Four-point transformation
- Homography
- getPerspectiveTransform()
- warpPerspective()

10. Image Warping

- warpAffine()
- warpPerspective()
- Remapping
- Coordinate transformations

LEVEL 4 — IMAGE PROCESSING

11. Image Filtering

- Convolution
- Kernels
- Averaging filter
- Gaussian blur
- Median blur
- Bilateral filtering
- Custom filters

12. Image Enhancement

- Brightness
- Contrast
- Histogram
- Histogram equalization
- CLAHE
- Gamma correction

13. Thresholding

- Binary thresholding
- Binary inverse
- Truncation
- Adaptive thresholding
- Otsu thresholding

14. Morphological Operations

- Structuring elements
- Erosion
- Dilation
- Opening
- Closing
- Morphological gradient
- Top hat
- Black hat

15. Edge Detection

- Image gradients
- Sobel
- Scharr
- Laplacian
- Canny
- Edge pipelines

LEVEL 5 — CONTOURS & SHAPE ANALYSIS

16. Contours

- What are contours?
- findContours()
- Contour hierarchy
- Contour retrieval modes
- Contour approximation

17. Contour Features

- Area
- Perimeter
- Centroid
- Bounding rectangle
- Rotated rectangle
- Minimum enclosing circle
- Convex hull
- Convexity defects

18. Shape Analysis

- Polygon approximation
- Shape matching
- Geometric properties
- Object counting
- Connected components

LEVEL 6 — FEATURE DETECTION & DESCRIPTION

19. Corner Detection

- Harris Corner Detector
- Shi-Tomasi
- Good Features to Track

20. Feature Descriptors

- Feature detection vs description
- SIFT
- ORB
- BRISK
- AKAZE

21. Feature Matching

- Brute Force Matcher
- BFMatcher
- FLANN
- Descriptor matching
- Ratio test
- Homography-based matching

LEVEL 7 — VIDEO PROCESSING

22. Video Input/Output

- VideoCapture
- VideoWriter
- Video files
- FPS
- Resolution
- Codec basics

23. Webcam Processing

- Camera capture
- Real-time frames
- Camera properties
- FPS measurement
- Real-time pipelines

24. Background Subtraction

- Background modeling
- MOG
- MOG2
- KNN
- Foreground masks

25. Optical Flow

- Motion estimation
- Lucas-Kanade
- Sparse optical flow
- Farneback
- Dense optical flow

LEVEL 8 — OBJECT TRACKING

26. OpenCV Tracking

- Tracking vs detection
- Single-object tracking
- Tracker initialization
- Bounding-box tracking

27. Tracking Algorithms

- CSRT
- KCF
- MIL
- MOSSE
- MedianFlow

28. Multi-Object Tracking

- Detection + tracking
- Track IDs
- Track management
- Association
- Basic SORT concepts

LEVEL 9 — CAMERA & CALIBRATION

29. Camera Model

- Pinhole camera model
- Coordinate systems
- Image coordinates
- Camera coordinates
- World coordinates

30. Camera Calibration

- Intrinsic parameters
- Extrinsic parameters
- Camera matrix
- Distortion coefficients
- Chessboard calibration
- Calibration images

31. Lens Distortion

- Radial distortion
- Tangential distortion
- Undistortion
- `undistort()`
- `initUndistortRectifyMap()`

32. Pose Estimation

- Perspective-n-Point
- `solvePnP()`
- Rotation vectors
- Translation vectors
- `Rodrigues()`
- Camera pose

LEVEL 10 — 3D VISION

33. Stereo Vision

- Stereo cameras
- Rectification
- Correspondence
- Disparity
- Depth estimation

34. Epipolar Geometry

- Epipolar lines
- Fundamental matrix
- Essential matrix
- Stereo geometry

35. Depth

- Disparity-to-depth
- Depth maps
- RGB-D
- Depth visualization

36. 3D Reconstruction

- Point clouds
- Triangulation
- 3D coordinates
- `triangulatePoints()`

LEVEL 11 — ADVANCED OPENCV

37. Hough Transform

- Hough Lines
- Probabilistic Hough Lines
- Hough Circles
- Shape detection

38. Image Pyramids

- Gaussian pyramid
- Laplacian pyramid
- Multi-scale processing

39. Template Matching

- Template matching
- Matching methods
- Thresholding results
- Multi-scale matching

40. Watershed & Segmentation

- Watershed algorithm
- Distance transform
- Marker-based segmentation

41. GrabCut

- Foreground extraction
- Background modeling
- Interactive segmentation

LEVEL 12 — OCR & DOCUMENT PROCESSING

42. Document Processing

- Document detection
- Perspective correction
- Document scanning
- Contour-based document extraction

43. OCR Pipeline

- Preprocessing
- Thresholding
- Noise removal
- Text region detection
- OCR integration

44. OpenCV + OCR Engines

- Tesseract integration
- EasyOCR integration
- PaddleOCR integration
- OCR post-processing

LEVEL 13 — OPENCV DNN

45. OpenCV DNN Module

- cv2.dnn
- Loading pretrained models
- Blob creation
- Forward inference
- Model outputs

46. Model Formats

- ONNX
- TensorFlow models
- Caffe
- Darknet
- Model conversion basics

47. DNN Inference

- Image classification
- Object detection
- Segmentation
- Preprocessing
- Postprocessing
- Confidence thresholds
- NMS

LEVEL 14 — PERFORMANCE & PRODUCTION

48. Performance Optimization

- ROI processing
- Vectorization
- Efficient NumPy usage
- Memory management
- FPS optimization
- Profiling

49. Parallel Processing

- Multithreading concepts
- Multiprocessing concepts
- Pipeline optimization
- Producer-consumer pipelines

50. GPU Acceleration

- CUDA basics
- OpenCV CUDA module
- GPU vs CPU processing
- GPU memory transfers

51. Real-Time Vision Pipelines

- Camera → preprocessing → inference → postprocessing
- Frame buffering
- FPS measurement
- Latency
- Throughput
- Real-time visualization

LEVEL 15 — ADVANCED OPENCV INTEGRATION

52. *OpenCV + NumPy*

- Efficient array operations
- Memory layout
- Vectorization
- Zero-copy concepts

53. *OpenCV + PyTorch*

- Tensor ↔ NumPy
- Image preprocessing
- Model input pipelines
- Model output processing

54. *OpenCV + ONNX Runtime*

- ONNX inference
- Preprocessing
- Postprocessing
- Runtime optimization

55. *OpenCV + TensorRT*

- TensorRT inference
- Engine loading
- Preprocessing
- Postprocessing
- Real-time inference